

Antigens

Definitions

Immunogen is any agent capable of inducing a specific immune response

Antigen is any agent reacts with the products of a specific immune response

Immunogenicity is the ability of an antigen to stimulate an immune response.

Hapten - A substance that is non-immunogenic but which can react with the products of a specific immune response. Haptens are small molecules which could never induce an immune response when administered by themselves but which can when coupled to a carrier molecule

- All immunogens are antigens, but not all antigens are immunogens.
- The structural portion of the antigen for binding antibodies or TCR is called as an **epitope** or **antigenic determinant**.
- **Antibody (Ab)**-A specific protein which is produced in response to an immunogen and which reacts with an antigen
- The typical size of an **epitope for ab** binding is about 5-7 amino acid residues. The region of the ab for binding an epitope is termed **antigen-binding site**.

- The size of an epitope for binding the corresponding TCR is generally larger of 10-15 amino acid residues. The aa residues of the antigenic peptide responsible for binding MHC molecules are called the **anchor residues**.
- Antigen may have single epitope or may have multiple epitopes of the same specificity (polysaccharides-**Polyvalency or Multivalency**); or combination of different epitopes on the same molecule (protein).

Factors influencing immunogenicity

A. Contribution of the Immunogen

1. Foreignness
2. Size
3. Chemical Composition
4. Physical form
5. Degradability

B. Contribution of the Biological System

- Genetic Factors
- Age

C. Method of Administration

- Dose
- Route
- Adjuvants

Chemical nature of immunogens

- A. Proteins
- B. Polysaccharides
- C. Nucleic Acids
- D. Lipids

Types of Antigens

A. T-independent Antigens –

- directly stimulate the B cells
- polysaccharides are T-independent antigens

Properties of T-independent antigens

a. **Polymeric structure** – the same antigenic determinant repeated many times

b. **Polyclonal activation of B cells** - Many of these antigens can activate B cell clones specific for other antigens (**polyclonal activation**)

c. Resistance to degradation - more resistant to degradation and thus they persist for longer periods of time and continue to stimulate the immune system.

B. T-dependent Antigens

- do not directly stimulate the production of antibody without the help of T cells
- Proteins are T-dependent antigens
- Structurally these antigens are characterized by a few copies of many different antigenic determinants

Antigenic Determinants Recognized by B cells and Ab

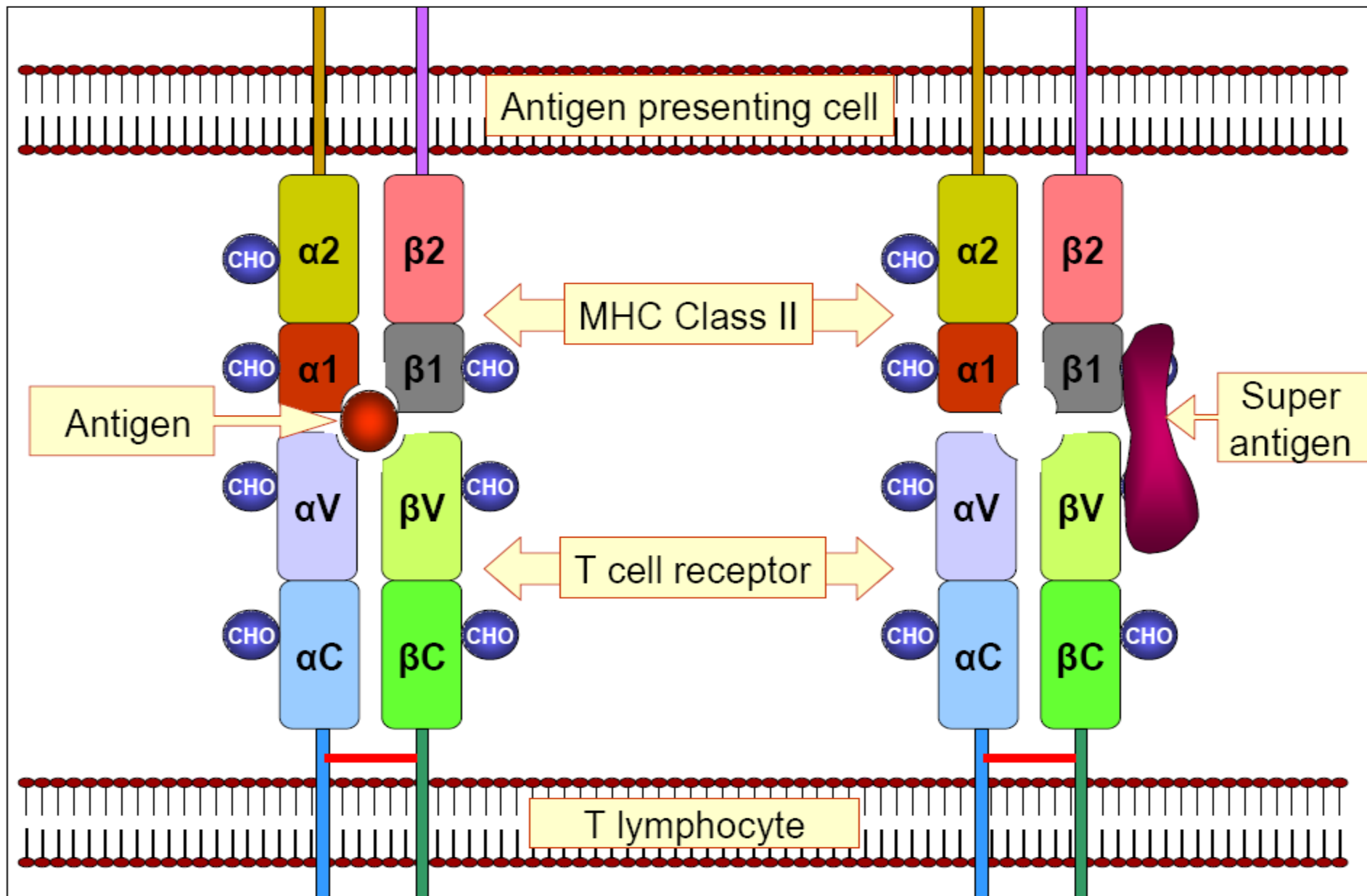
- Composition
 - Proteins, polysaccharides, nucleic acids
 - Sequence (linear) determinants
 - Conformational determinants
- Size
 - 4-8 residues
- Number
 - Limited (immunodominant epitopes)
 - Located on the external surfaces of the Ag

Antigenic Determinant Recognized by T cells

- Composition
 - Proteins (some lipids)
 - Sequence determinants
 - Processed
 - MHC presentation (lipid presentation by MHC-like CD1)
- Size
 - 8 -15 residues
- Number
 - Limited to those that can bind to MHC

Superantigens

- Conventional T-dependent antigen, only a small fraction (1 in 10^4 - 10^5) of the T cell population is able to recognize the antigen and become activated (monoclonal/oligoclonal response).
- However, there are some antigens which polyclonally activate a large fraction of the T cells (up to 25%)- **Superantigens**



- Examples of superantigens include:
 - Staphylococcal enterotoxins (food poisoning),
 - Staphylococcal toxic shock toxin (toxic shock syndrome),
 - Staphylococcal exfoliating toxins (scalded skin syndrome)
 - Streptococcal pyrogenic exotoxins (shock).
- Although the bacterial superantigens are the best studied there are superantigens associated with viruses and other microorganisms as well.
- The diseases associated with exposure to superantigens are, in part, due to hyper activation of the immune system and subsequent release of biologically active cytokines by activated T cells.

Antigen processing and presentation

- **Antigen Processing** – is a metabolic process that digests proteins into peptides, which can then be displayed on the cell membrane together with MHC class I or II molecules
- **Endogenous antigens**- peptides derived from antigens that have been processed within the cytoplasm of the cell
- **Exogenous antigens**- Peptides derived from antigens that are internalized by phagocytosis /endocytosis and processed within the endocytic pathway
- T cells recognize Ag in the form of short peptides in the context of either MHC class-I or MHC class-II molecules

- The two types of T cells (CD4+ and CD8+) recognize Ag bound and presented by two distinct MHC classes:

CD4+ T cells recognize Ag in the context of MHC class-II while CD8+ T cells recognize Ag in the context of MHC class-I molecules

- This dichotomy is imposed by the fact that the CD4 and CD8 molecules ligate (bind to) the MHC class-II and the MHC class-I molecules, respectively.

The CD4 and CD8 molecules are called **co-receptors**. The CD4 molecule binds to the outer surface of the $\beta 2$ domain of the MHC class-II molecule.

The CD8 molecule makes contact with the $\alpha 3$

- Strictly speaking, all cells expressing MHC class I or class II molecules can present peptides. However, only cells presenting peptides associated with MHC class II molecules to CD4 T cells are called **antigen-presenting cells (APC)**. Cells presenting peptides with MHC class I molecules are called **target cells**.

There are two types of APC:

A. Professional APC

- **Dendritic cells**: Are the most effective APC. They express constitutively high levels of MHC class II molecules and co-stimulatory molecules.

- **Macrophages**: Need to be activated by phagocytosis of microorganisms before they express MHC class II molecules or co-stimulatory molecules.
- **B cells**: Express MHC class II molecules constitutively but need to be activated before they express co-stimulatory molecules.

B. **Nonprofessional APC**

These are cells that can serve as APC for short periods of time during inflammation.

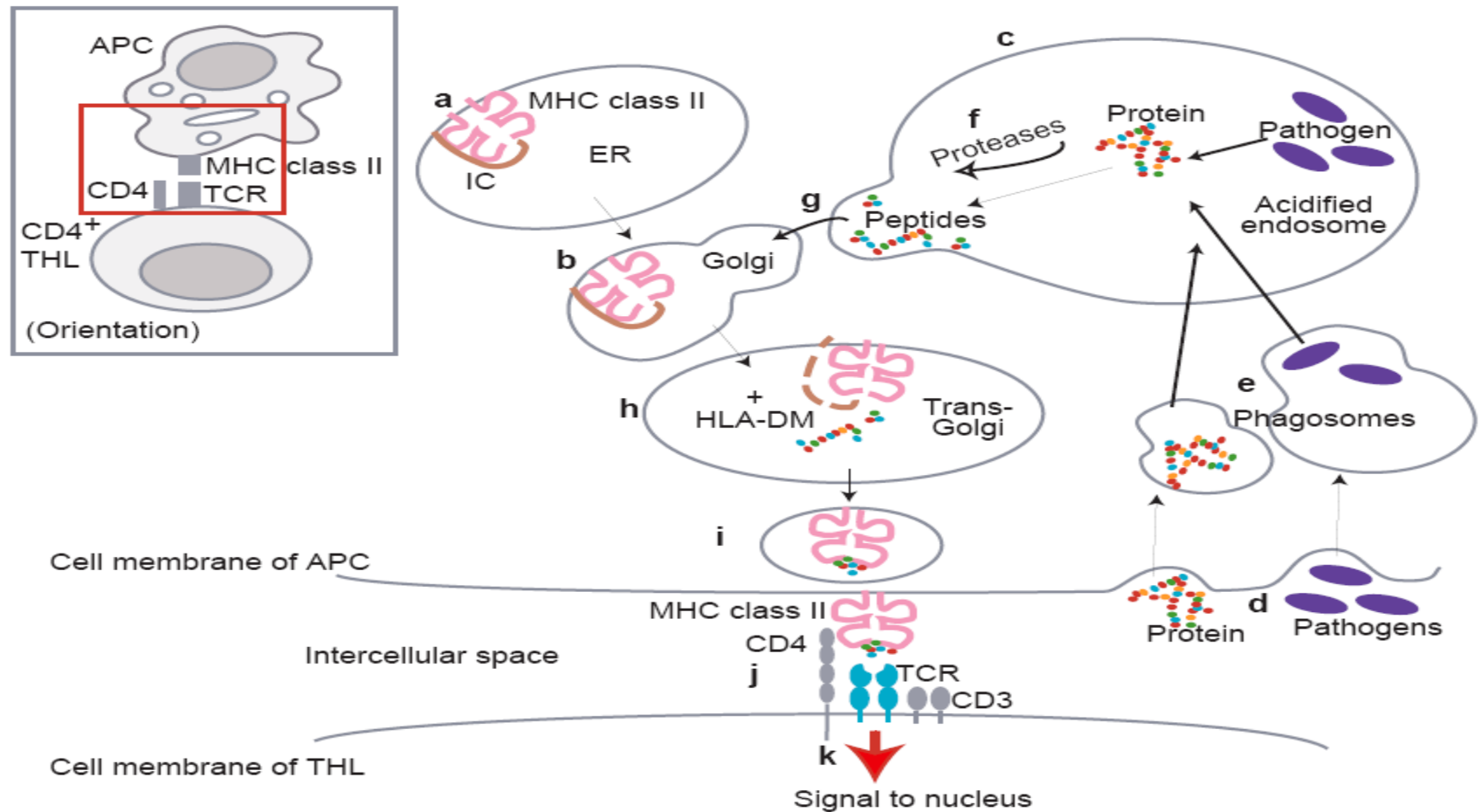
Eg. **epithelial cells and endothelial cells.**

Table 5-3 Properties and Functions of Antigen-Presenting Cells

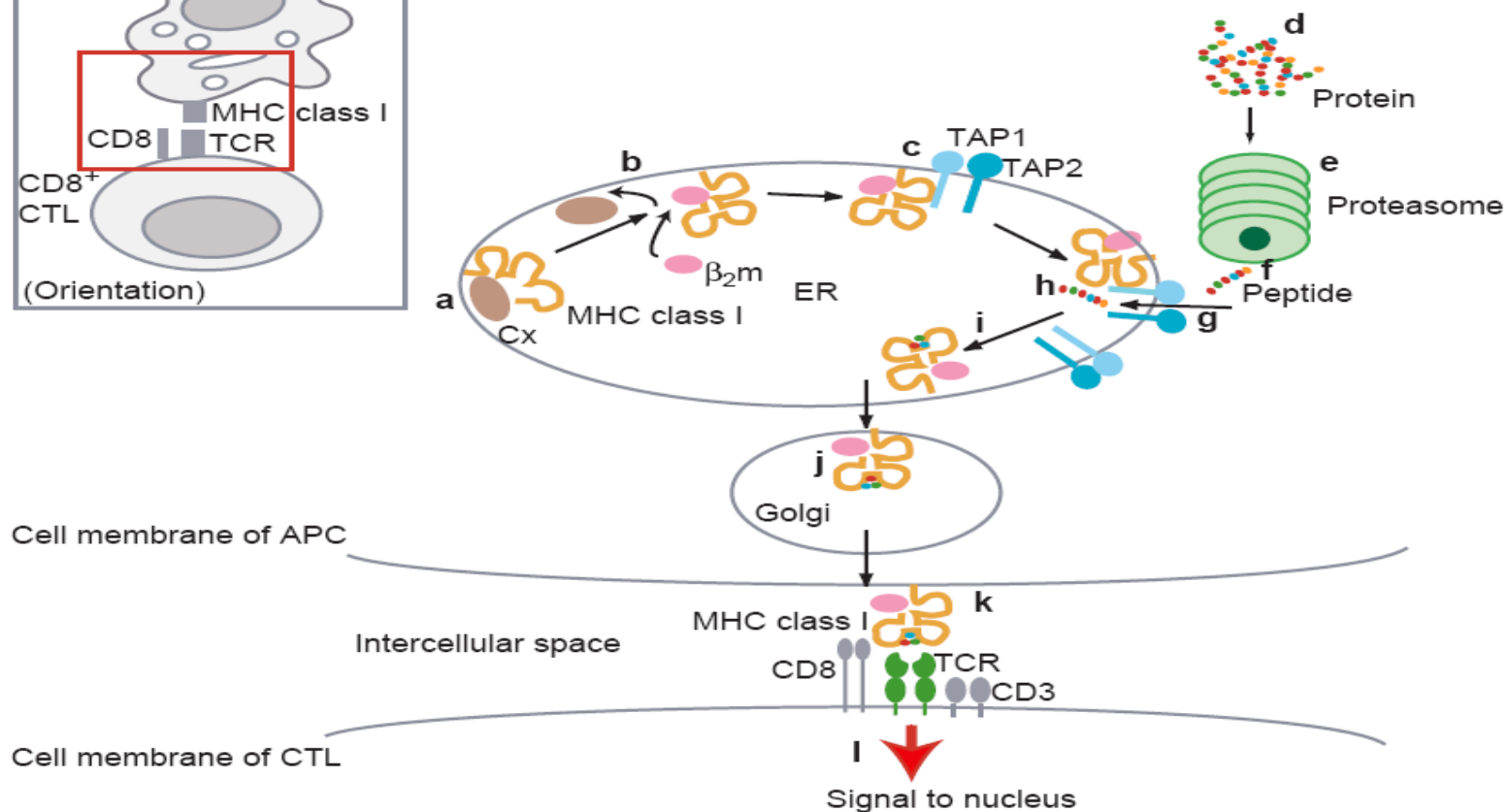
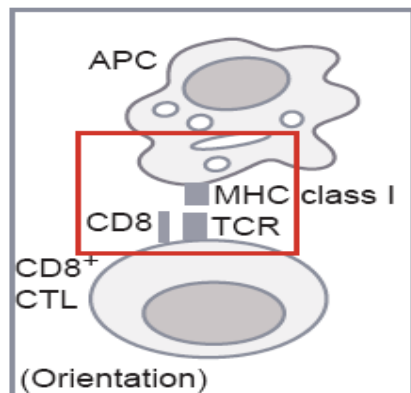
Cell Type	Expression of		Principal Function
	Class II MHC	Costimulators	
Dendritic cells (Langerhans cells)	Constitutive; increases with maturation; increased by IFN- γ	Constitutive; increases with maturation; inducible by IFN- γ , CD40-CD40L interactions	Initiation of T cell responses to protein antigens (priming)
Macrophages	Low or negative; inducible by IFN- γ	Inducible by LPS, IFN- γ , CD40-CD40L interactions	Induction and effector phase of cell-mediated immune responses
B lymphocytes	Constitutive; increased by IL-4	Inducible by T cells (CD40-CD40L interactions), antigen receptor cross-linking	Antigen presentation to CD4 ⁺ helper T cells in humoral immune responses (cognate T cell-B cell interactions)
Vascular endothelial cells	Inducible by IFN- γ ; constitutive in humans	Constitutive (inducible in mice)	May promote activation of antigen-specific T cells at site of antigen exposure
Various epithelial and mesenchymal cells	Inducible by IFN- γ	Probably none	No known physiologic function

Abbreviations: IFN- γ , interferon- γ ; IL-4, interleukin-4; LPS, lipopolysaccharide

- The APCs process Ag along **two distinct pathways** depending to how the Ag/pathogen has gained access and where it is localized within the cell:
 1. The **cytosolic (Endogenous) pathway**
 2. The **vesicular pathway (Endocytic) pathway**
- Pathogens replicating in the cytosol are processed and presented to CD8+ T cells in the context of MHC class-I molecules while intravesicular pathogens are processed and presented to CD4+ T cells in the context of MHC class-II molecules



Degradation and transport of antigens that bind major histocompatibility complex (MHC) class II molecules
Expert Reviews in Molecular Medicine



Degradation and transport of antigens that bind major histocompatibility complex (MHC) class I molecules

Expert Reviews in Molecular Medicine

- Group 1 CD1 molecules mainly present lipid antigens to clonally diverse T cells that mediate adaptive immunity to the vast range of microbial lipid antigens. By contrast, CD1d (group 2) molecules present lipid antigens to natural killer T (NKT) cells
- CD1 proteins survey the endocytic pathway to intersect and bind lipid antigens
- Saposins mediate the loading of lipids onto CD1 molecules in lysosomes

The Major Histocompatibility Complex (MHC)

- Major Histocompatibility Complex

Cluster of genes found in all mammals

Its products play role in discriminating self/non-self

Participant in both humoral and cell-mediated immunity

- MHC act as antigen presenting structures

- In Human MHC is found on Chromosome 6

referred to as HLA complex

- In Mice MHC is found on Chromosome 17

referred to as H-2 complex

The MHC Genes

- Genes Of MHC Organized In 3 Classes
 - Class I MHC genes
 - Glycoproteins expressed on all nucleated cells
 - Major function to present processed Ags to T_C
 - Class II MHC genes
 - Glycoproteins expressed on $M\Phi$, B-cells, DCs
 - Major function to present processed Ags to T_H
 - Class III MHC genes
 - Products that include secreted proteins that have immune functions. Ex. Complement system, inflammatory molecules

- Class I MHC Genes Found In Regions A, B and C In Humans (K and D In Mice)
- Class II MHC Genes Found In Regions DR, DP and DQ (IA and IE In Mice)
- Class I and Class II MHC Share Structural Features
 - Both involved in APC
- Class III MHC Have No Structural Similarity To Class I and II
 - Ex. TNF, heat shock proteins, complement components

Mouse H-2 complex

Complex	H-2						
MHC class	I	II		III		I	
Region	K	IA	IE	S		D	
Gene products	H-2K	IA $\alpha\beta$	IE $\alpha\beta$	C' proteins	TNF- α TNF- β	H-2D	H-2L

Human HLA complex

Complex	HLA							
MHC class	II			III		I		
Region	DP	DQ	DR	C4, C2, BF		B	C	A
Gene products	DP $\alpha\beta$	DQ $\alpha\beta$	DR $\alpha\beta$	C' proteins	TNF- α TNF- β	HLA-B	HLA-C	HLA-A

The MHC is Polygenic and Polymorphic

MHC is polygenic because it has several genes for each class (3 for class-I, 3 for class-II).

MHC is polymorphic due to the relatively large number of alleles within each MHC class.

MHC Polymorphism

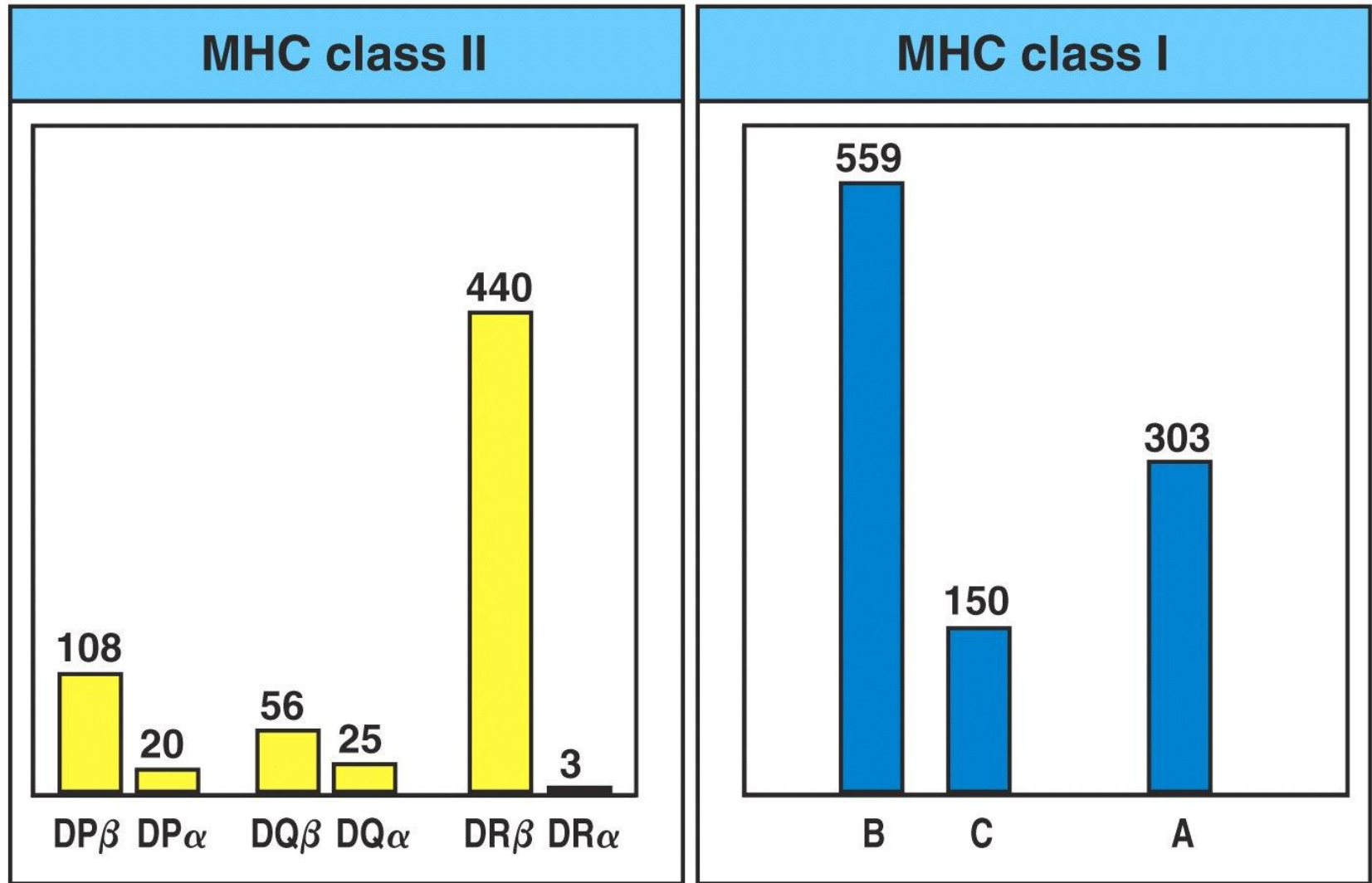


Figure 5-13 Immunobiology, 6/e. (© Garland Science 2005)

Consequences of MHC polymorphism

The combined effect of polymorphism and polygeny is the generation of a highly diverse repertoire of MHC molecules capable of presenting a large variety of Ag peptides to T cells.

MHC Classification and structure

The two types of polymorphic genes, namely the class I and II encodes two groups of structurally distinct but homologous proteins.

MHC class-I is made up of two polypeptide chains: One large called α and one small called β_2 microglobulin.

The α chain has 3 extracellular domains: α_1 , α_2 and α_3 , a transmembrane and a cytoplasmic segment.

MHC Classification and structure

The $\alpha 1$ and $\alpha 2$ domains form a cleft where the Ag fragment binds.

The $\beta 2$ microglobulin has only one extracellular domain

- No transmembrane portion

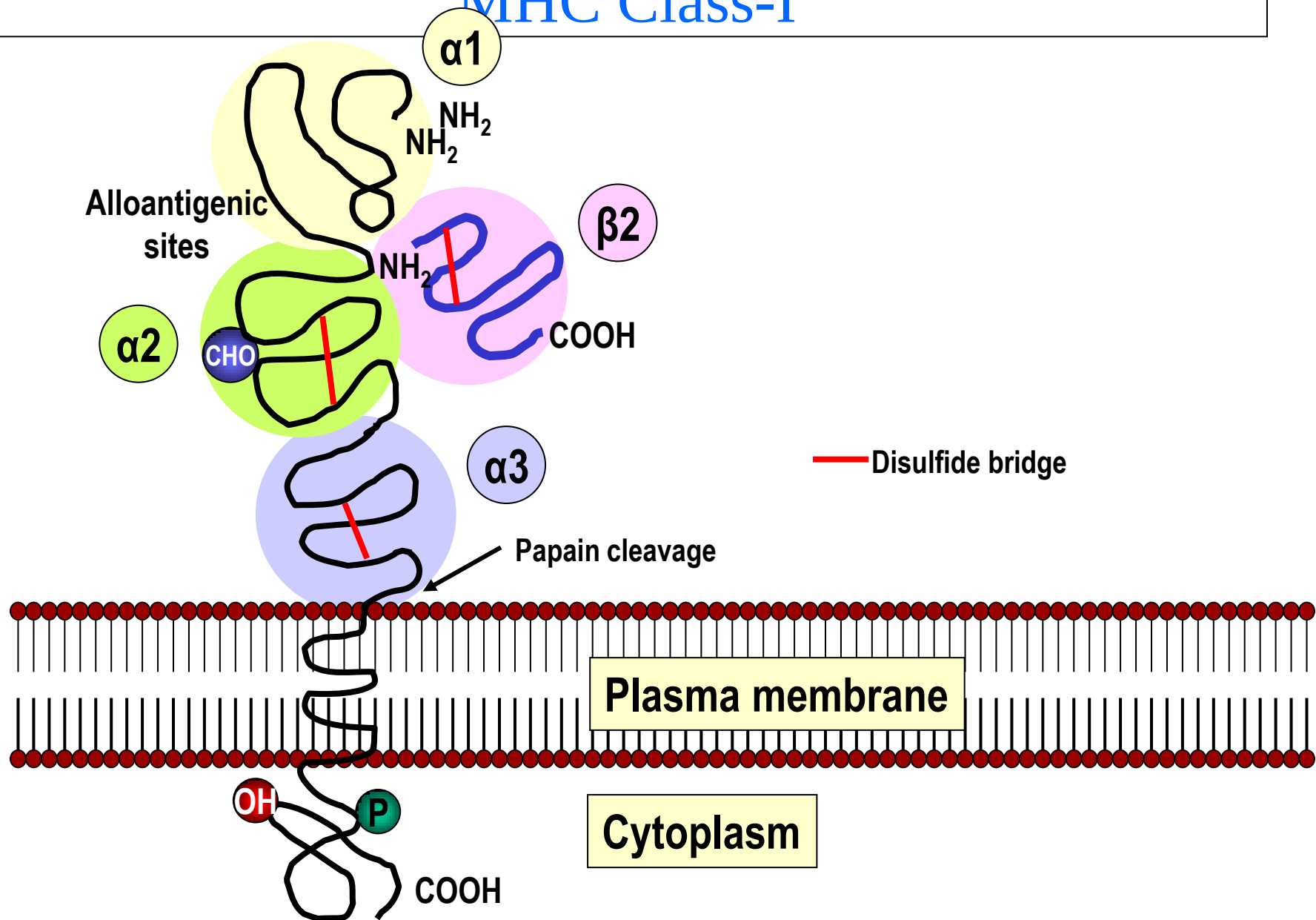
- No cytoplasmic tail

- Binds to the $\alpha 3$ domain non-covalently

Immunoglobulin-like region – highly conserved

$\alpha 3$ domain - site to which CD8 on T cell binds

MHC Class-I



MHC Class-II

Composed of two polypeptide chains α and β .

Each chain has two extracellular domains $\alpha 1$ $\alpha 2$ and $\beta 1$ and $\beta 2$

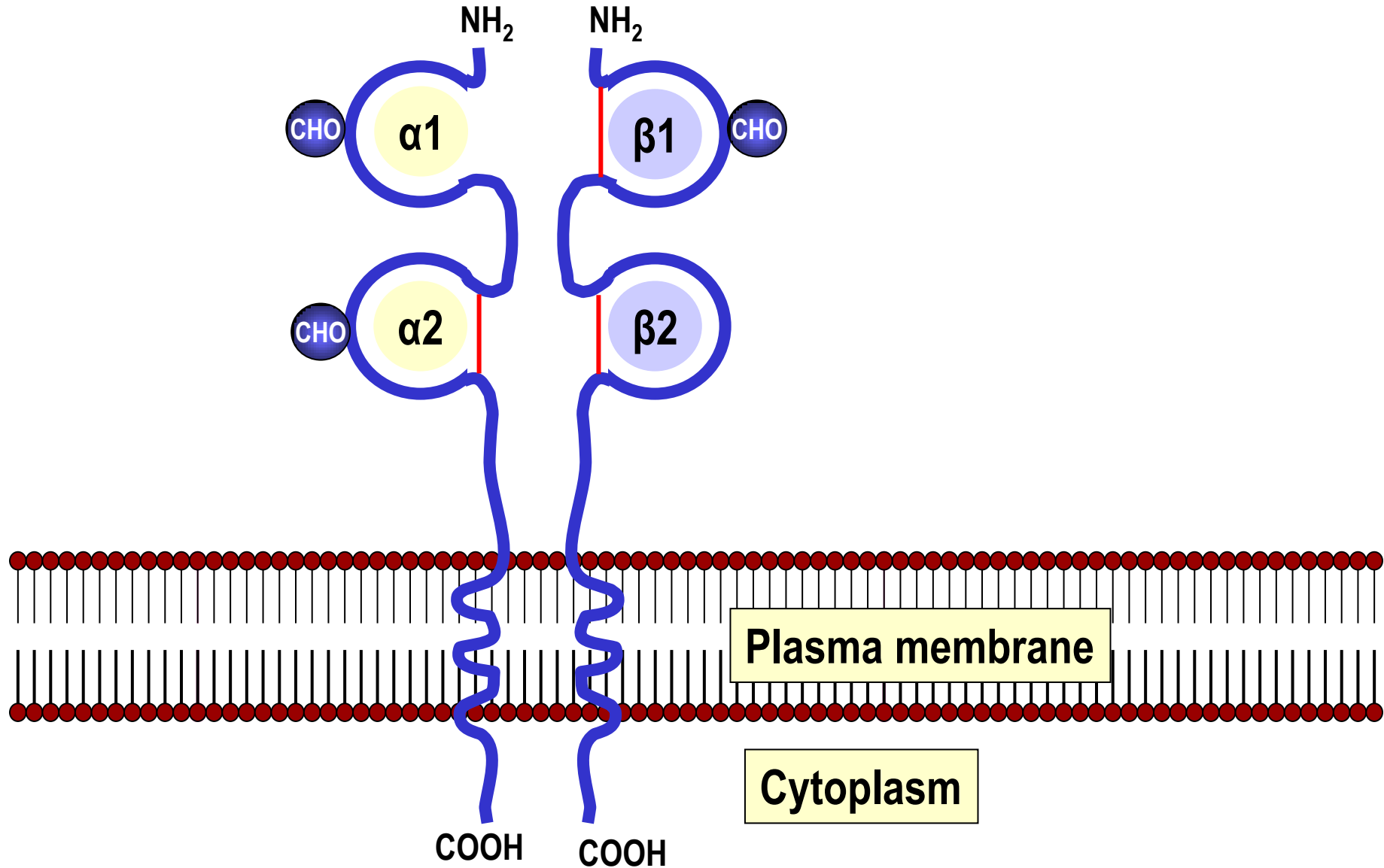
A transmembrane segment

A cytoplasmic tail.

The $\alpha 1$ and $\beta 1$ domains form the cleft for the Ag fragment

Immunoglobulin-like region – conserved $\alpha 2$ and $\beta 2$ domains – $\beta 2$ is site to which CD4 on T cell binds

The MHC class-II



Class I or II MHC Molecules and Interaction between Peptides

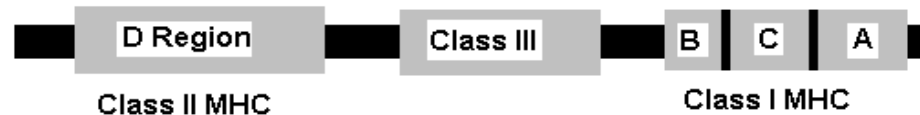
- MHC molecules show a broad specificity
- Each MHC molecule has a single peptide binding cleft
- The peptide sequences that bind to MHC molecules are distinct from those recognized by T cells
- The affinity of peptide-MHC interactions is much lower than that of antigen-antibody binding
- Peptide-MHC complexes persist long due to low rate of dissociation
- MHC molecules don't discriminate between self and non Self antigens

- The binding of peptides to MHC molecules is a non-covalent interaction mediated by residues both in the peptides and in the clefts of the MHC molecules
- Amino acids from both the antigenic peptide and the MHC molecules contribute to T cell antigen recognition, with the peptide being responsible for the fine specificity and the MHC residues accounting for the MHC restriction of the T cells

Genomic Organization of MHC I and II

- In Humans, the MHC is located on the short arm of chromosome 6 and β -2 Microglobulin is encoded by a gene on chromosome 15
- Human MHC occupies 3500kb
- Class I genes are in the most telomeric region and class II are in the most centromeric region of HLA locus
- Other genes in the class II locus
 - TAP 1 and 2
 - subunits of a cytosolic protease complex (Proteasome)
 - HLA-DM

- MHC class III = Between class I and II gene
 - Complement factors C2, C4, & Factor B
 - Genes encoding in this MHC III do not play any role in antigen presentation



Expression of MHC complex Molecules

- Class I expressed on all nucleated cells
- Class II expressed on APCs (Macrophages, B cells and Dendritic cells)
- Expression of MHC molecules is increased by cytokines produced during innate and adaptive immune responses
 - Interferon α , β and γ
 - Tumor necrosis factor and lymphotoxin
 -  increases expression of class I molecules
 - Interferon γ stimulates expression of class II molecules

Peptide-binding grooves for class I and class II MHC are structurally similar

- Both have a peptide-binding groove with a wall of two α helices and a floor of eight β -pleated sheets
- Close-ended groove for class I MHC requires an 8-10 amino acid-length peptide to bind; open-ended groove for Class II MHC lets it bind a peptide 13-25 amino acids long

Aspects of MHC

1. MHC molecules are membrane-bound.
Recognition by T cells requires cell-cell contact.
2. Peptide from cytosol associates with class I MHC and is recognized by Tc cells.
Peptide from vesicles associates with class II MHC and is recognized by Th cells.

3. Mature T cells must have a T cell receptor that recognizes the peptide associated with MHC.
4. Each MHC molecule has only one binding site. The different peptides a given MHC molecule can bind all to the same site, but only one at a time.

6. MHC polymorphism is determined only in the germline. There are no recombinational mechanisms for generating diversity.
7. Because each MHC molecule can bind many different peptides, binding is termed degenerate.
8. Cytokines (especially interferon- γ) increase level of expression of MHC.

Reading Assignment

- The **cytosolic (Endogenous) pathway**
- The **vesicular pathway (Endocytic) pathway**